

Sensitivity test - Window size

[ODSA_README](#)

 [ODSA-AIM_and_METHOD](#)

The current setup

Fixed **50 km rectangular window with 10 km step**. The Lomb-Scargle PSD is computed per-window, then averaged. Peaks are detected in the whitened residuals after power-law fitting.

✓ ~~Test 4 window sizes (e.g., 30 km, 50 km, 75 km, 100 km)~~

WHY

`min_freq = 1 / window_size` directly controls what wavelengths the code can resolve.

At 50 km, we cannot see anything longer than 50 km. Changing window size shifts this cutoff and changes the number of windows averaged (affecting noise reduction). **This could meaningfully change the confirmed wavelength counts.**

Findings summary

Metric	30 km	50 km	75 km	100 km
Trajectories	20	20	20	20
Segments	38	38	38	38
β exponent	2.2, [1.3, 2.8]	2.2, [1.3, 2.8]	2.2, [1.3, 2.8]	2.2, [1.3, 2.8]
ζ exponent	0.6, [0.2, 0.9]	0.6, [0.2, 0.9]	0.6, [0.2, 0.9]	0.6, [0.2, 0.9]
Vertical Relief	1311.3m	1311.3m	1311.3m	1311.3m
Bed elevation range	[-1874.8, 1225.2]	[-1874.8, 1225.2]	[-1874.8, 1225.2]	[-1874.8, 1225.2]
Ice thickness	2990.6m	2990.6m	2990.6m	2990.6m
flow orientation	18 $\parallel$ /14 $\perp$ /6 $\circ$	18 $\parallel$ /14 $\perp$ /6 $\circ$	18 $\parallel$ /14 $\perp$ /6 $\circ$	18 $\parallel$ /14 $\perp$ /6 $\circ$
wavelength	~	~	~	~
candidates	4, [10777, 13772]	4, [10777, 13772]	4, [10777, 13772]	4, [10777, 13772]
confirmed count	1327	1456	1764	1908
confirmed mean	697.1 m	1068.5 m	1228.0 m	1405.3 m
confirmed range	[53.5, 19692.3] m	[54.2, 22072.0] m	[54.2, 37289.6] m	[54.2, 57529.3] m
exported windows	227	229	181	141

Everything that's segment-level is identical across all runs

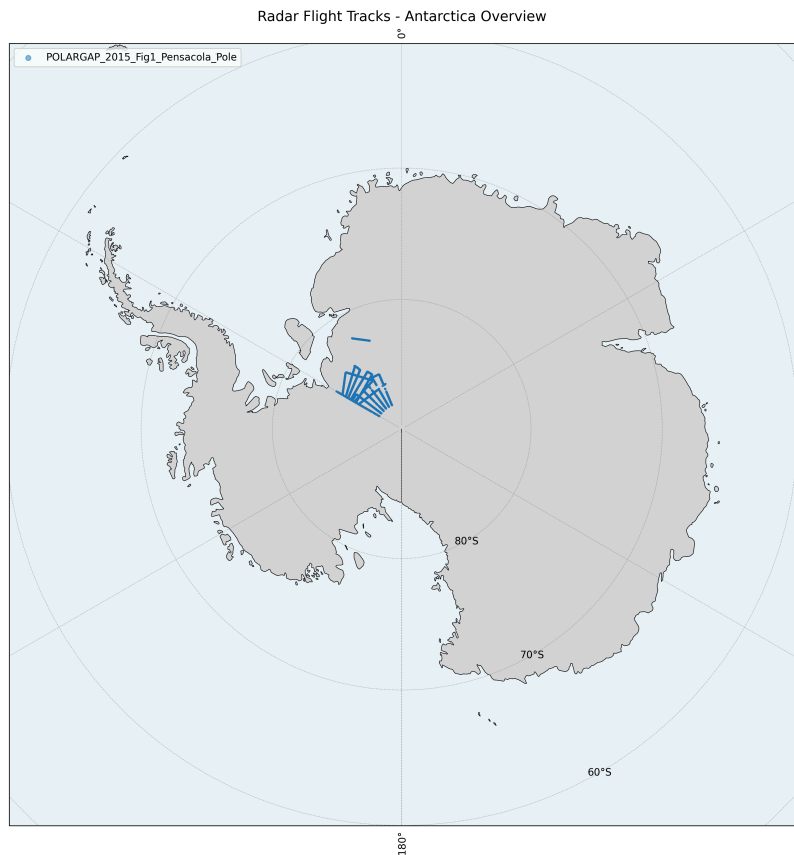
The analysis is insensitive to window size. **The 50 km default is adequate.**

NOTE: Since the candidate wavelengths (4, same range) didn't change at all with window size, this suggests that the peak detection isn't being fooled by side lobes, meaning that **Step2** in [TODO](#) is not very high priority.

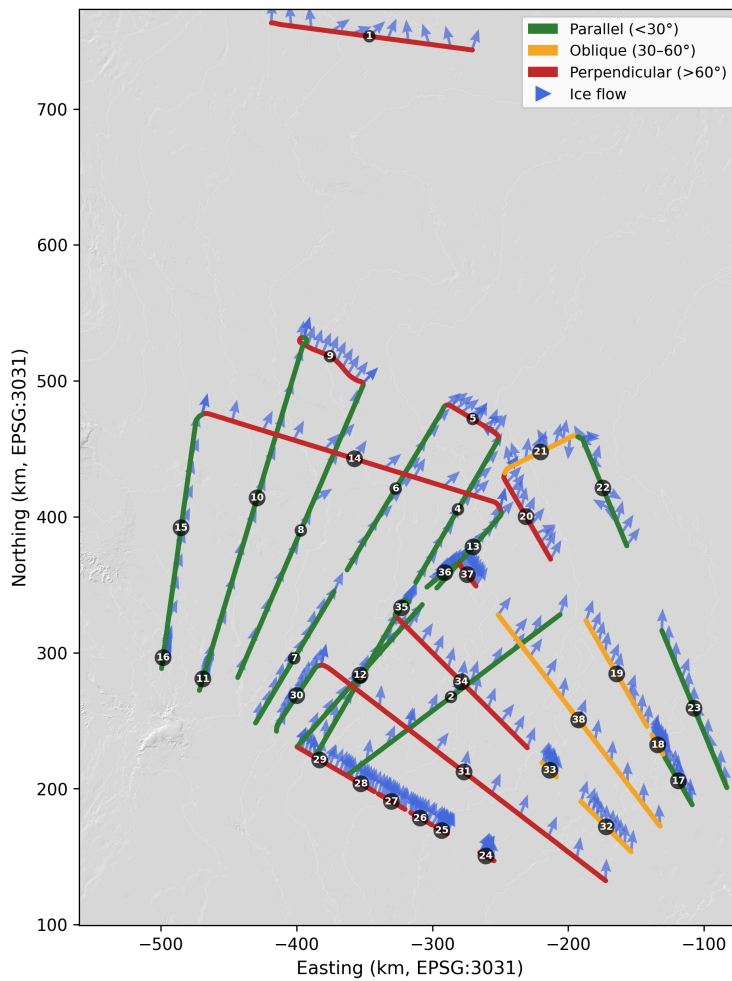
	30 km	50 km	75 km	100 km
Segment fits				
Linear regression	slope=0.0008 R²=0.006 p=6.35e-01	slope=0.0008 R²=0.005 p=6.59e-01	slope=0.0002 R²=0.000 p=9.24e-01	slope= 0.0003 R²=0.001 p=8.85e-01
cos²θ	$\beta_{\parallel} = 2.154 \pm 0.068$ $\beta_{\perp} = 2.218 \pm 0.075$ $\Delta\beta = -0.064$ R²=0.0087	$\beta_{\parallel} = 2.126 \pm 0.073$ $\beta_{\perp} = 2.193 \pm 0.081$ $\Delta\beta = -0.067$ R²=0.0081	$\beta_{\parallel} = 2.174 \pm 0.079$ $\beta_{\perp} = 2.201 \pm 0.087$ $\Delta\beta = -0.026$ R²=0.0011	$\beta_{\parallel} = 2.175 \pm 0.079$ $\beta_{\perp} = 2.204 \pm 0.087$ $\Delta\beta = -0.030$ R²=0.0014
Window fit				
Linear regression	slope=0.0007 R²=0.0044 p=2.6910e-01	slope=0.0003 R²=0.0009 p=6.4876e-01	slope=0.0006 R²=0.0059 p=3.0430e-01	slope=0.0011 R²=0.0206 p=8.9695e-02

TESTING DATA SET: Pensacola Pole Basin (PPB)

Radar tracks: BAS_2015_POLARGAP_AIR_BM3



Ice Flow Orientation: POLARGAP_2015_Fig1_Pensacola_Pole
P32.6



Segment List

#	Angle	km
1	65°	149.4
2	29°	196.7
3	9°	133.9
4	9°	123.3
5	65°	47.6
6	11°	139.8
7	11°	112.5
8	10°	235.7
9	75°	61.8
10	7°	252.8
11	2°	17.4
12	12°	139.5
13	10°	76.3
14	72°	232.1
15	12°	171.2
16	14°	17.1
17	28°	40.5
18	40°	16.5
19	39°	89.9
20	61°	72.7
21	56°	61.4
22	28°	88.8
23	23°	125.1
24	76°	12.1
25	80°	10.2
26	72°	18.0
27	72°	22.6
28	75°	24.0
29	76°	38.9
30	22°	60.1
31	72°	263.0
32	51°	52.3
33	57°	15.0
34	69°	138.0
35	19°	14.2
36	25°	30.5
37	72°	20.2
38	53°	196.0

ODSA (**stratified**) measured roughness of each flight line track by orientation
The majority of segments were found to be parallel (n=18) and perpendicular (n=14).
while only 6 segments were found to be oblique.

RESULTS

50 km window

Terminal

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=== POLARGAP_2015_Fig1_Pensacola_Pole SUMMARY ===
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RESULTS SUMMARY (20 trajectories aggregated)  
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VERTICAL RELIEF (Max-Min):

-> Mean: 1311.3m | Range: [442.2, 2437.3]m

ABSOLUTE BED ELEVATION:

-> Min: -1874.8m | Max: 1225.2m (across all trajectories)

AVG SEGMENT LENGTH:

-> Mean: 114066.9m | Range: [20970.2, 262973.5]m

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POWER LAW EXPONENT (Beta):
-> Mean: 2.2 | Range: [1.3, 2.8]
HURST EXPONENT:
-> Mean: 0.6 | Range: [0.2, 0.9]
MEAN ICE THICKNESS:
-> Mean: 2990.6m | Range: [1801.3, 3821.0]m
.....
FLOW ORIENTATION (All Segments):
-> Oblique: 6 segments (15.8%)
-> Perpendicular: 14 segments (36.8%)
-> Parallel: 18 segments (47.4%)
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
-> Count: 1456
-> Mean: 1068.5m | Range: [54.2, 22072.0]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
-> Count: 4
-> Range: [10777m, 13772m]
.....
=====
Exported 229 window rows and 38 segment rows

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Observations

IFPA categorised the region containing these radar tracks as **Alpine (subglacial) / Selective erosion**. While the ODSA results indicate this region has a mean β value of 2.2 with a large range: [1.3, 2.8]. Putting the result between the **hard bed** ("Brownian": complex and rough) and the **sedimentary Infill** category ($\beta > 2.5$). This means that ODSA would classify this area as a combination of high relief terrain with some smoother areas.

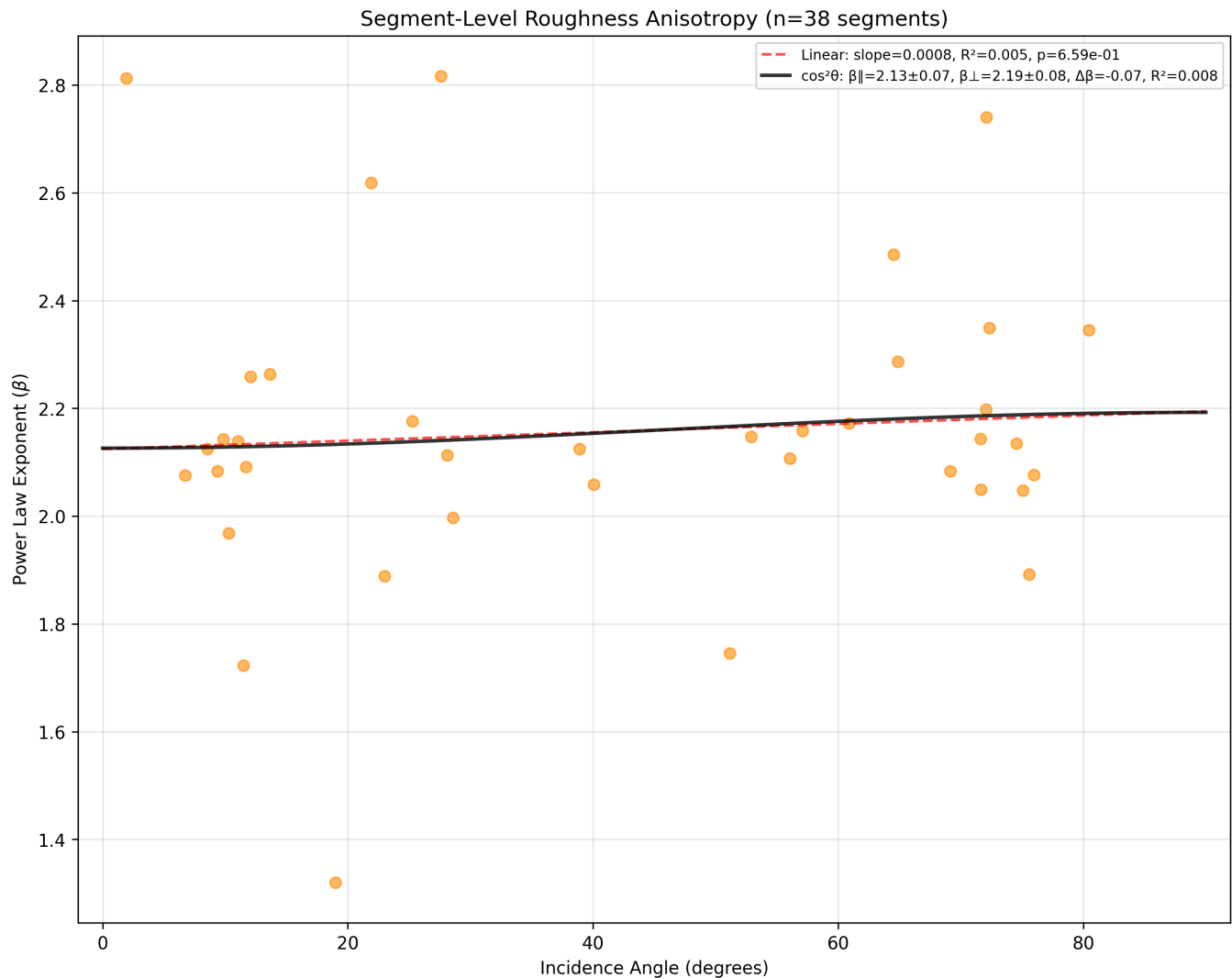
The mean value for the vertical relief of 1311.3 m | Range: [442.2, 2437.3]m

The height difference between the highest and lowest points available in the data for this area are Min: -1874.8 m | Max: 1225.2 m (across all trajectories).

The mean confirmed wavelength of 1068.5 m with the range going up to ~22 km indicates the roughness is dominated by structures at roughly the km-scale.

Anisotropy

Segment-level Roughness FIT



The $\cos^2\theta$ fit: $\beta_{\parallel} = 2.13 \pm 0.07$, $\beta_{\perp} = 2.19 \pm 0.08$, $\Delta\beta = -0.07$, $R^2=0.008$

The error bars of the parallel and perpendicular beta parameters **overlap** substantially:

2.13 ± 0.07 gives [2.06, 2.20], and 2.19 ± 0.08 gives [2.11, 2.27], this means that parallel and perpendicular roughness are indistinguishable

The anisotropy signal $\Delta\beta = -0.07$

Its significance can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.07}{\sqrt{0.07^2 + 0.08^2}} \approx \frac{0.07}{0.106} \approx 0.66$

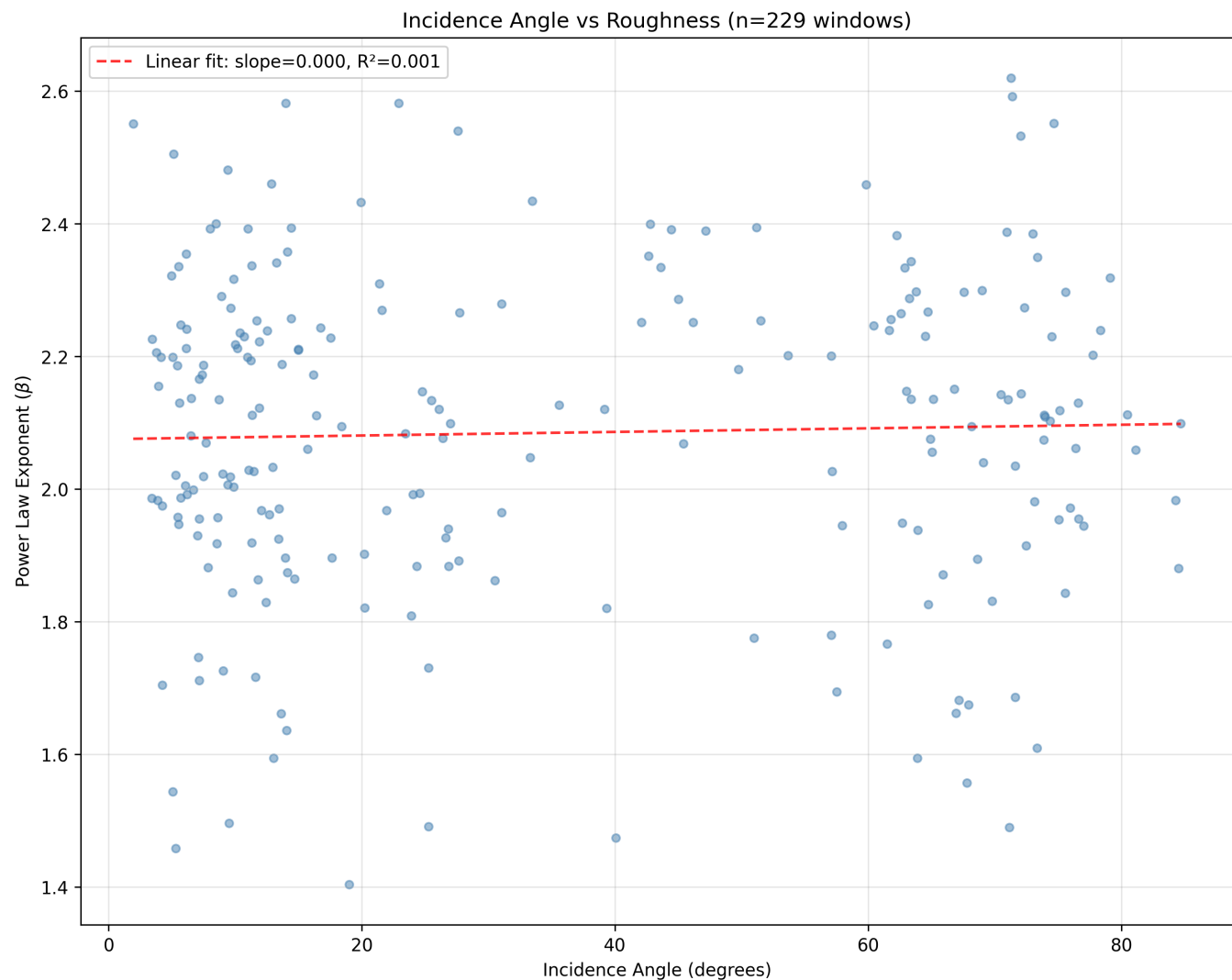
This is below 1.96 (95% confidence interval), which means that the result is not significant.

The $R^2=0.008$ implies that the model explains only 0.8% of variance. **Essentially no directional dependence detected.** This provides support for the **alpine** landscape classification (**isotropic landscape**) and rules out **selective erosion**.

The linear fit at the segment level has **slope=0.0008**, **$R^2=0.005$** , **$p=6.59e-01$**

Since **$p < 0.05$** means significant. At 0.659 this result is non-significant (there's a 66% chance you'd see this slope (or steeper) from pure noise). This means that the slope is essentially zero.

Window-level Roughness FIT



The window level analysis show a near-flat line fit (small slope, very modest R^2). That means that there is no systematic dependence of roughness on flow orientation, this is consistent with a non-streamlined landscape providing support for the **Alpine landscape** classification.

Conclusion:

PPB appears to be an Alpine region based on its isotropy and its bedrock hardness characteristics.

30 km

Terminal

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==== POLARGAP_2015_Fig1_Pensacola_Pole SUMMARY ====
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RESULTS SUMMARY (20 trajectories aggregated)
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VERTICAL RELIEF (Max-Min):
-> Mean: 1311.3m | Range: [442.2, 2437.3]m
```

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ABSOLUTE BED ELEVATION:
-> Min: -1874.8m | Max: 1225.2m (across all trajectories)
AVG SEGMENT LENGTH:
-> Mean: 114066.9m | Range: [20970.2, 262973.5]m
POWER LAW EXPONENT (Beta):
-> Mean: 2.2 | Range: [1.3, 2.8]
HURST EXPONENT:
-> Mean: 0.6 | Range: [0.2, 0.9]
MEAN ICE THICKNESS:
-> Mean: 2990.6m | Range: [1801.3, 3821.0]m
.....
FLOW ORIENTATION (All Segments):
-> Perpendicular: 14 segments (36.8%)
-> Oblique: 6 segments (15.8%)
-> Parallel: 18 segments (47.4%)
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
-> Count: 1327
-> Mean: 697.1m | Range: [53.5, 19692.3]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
-> Count: 4
-> Range: [10777m, 13772m]
.....
=====
Exported 277 window rows and 38 segment rows

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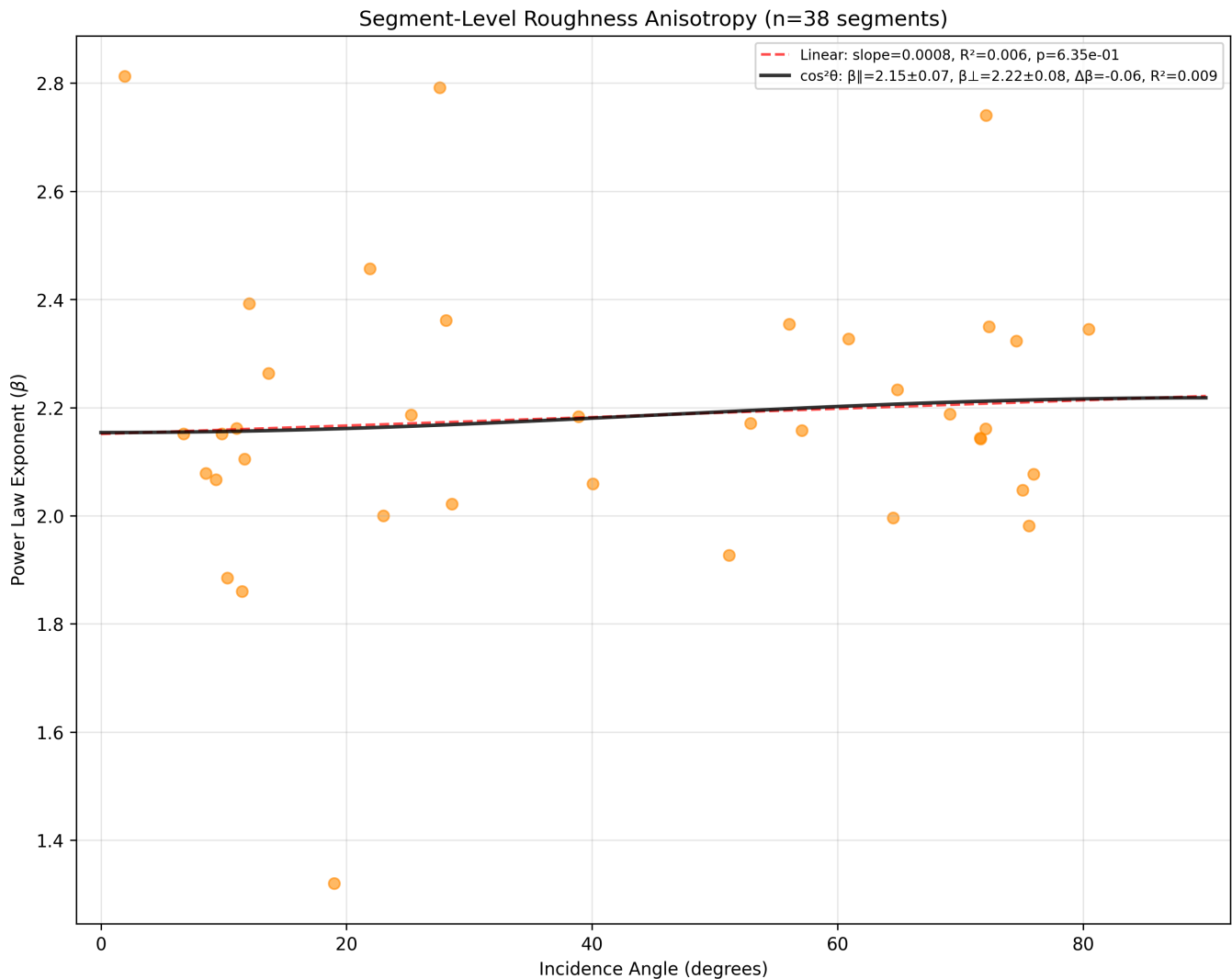
Observations

Mean β value and range, mean vertical relief and range, aggregated height minimum and maximum values are all IDENTICAL to the 50km window condition.

The mean confirmed wavelength of 697.1 m with the range going up to ~20 km indicates the roughness is dominated by structures at roughly the sub-km-scale.

Anisotropy

Segment-level Roughness FIT



The $\cos^2 \theta$ fit: $\beta_{\parallel} = 2.15 \pm 0.07$, $\beta_{\perp} = 2.22 \pm 0.08$, $\Delta\beta = -0.06$, $R^2=0.009$

The error bars of the parallel and perpendicular beta parameters **overlap** substantially:

2.15 ± 0.07 gives [2.08, 2.20], and 2.22 ± 0.08 gives [2.14, 2.30], this means that parallel and perpendicular roughness are indistinguishable.

The anisotropy signal $\Delta\beta = -0.06$

Its significance can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.06}{\sqrt{0.07^2 + 0.08^2}} \approx \frac{0.06}{0.106} \approx 0.56$

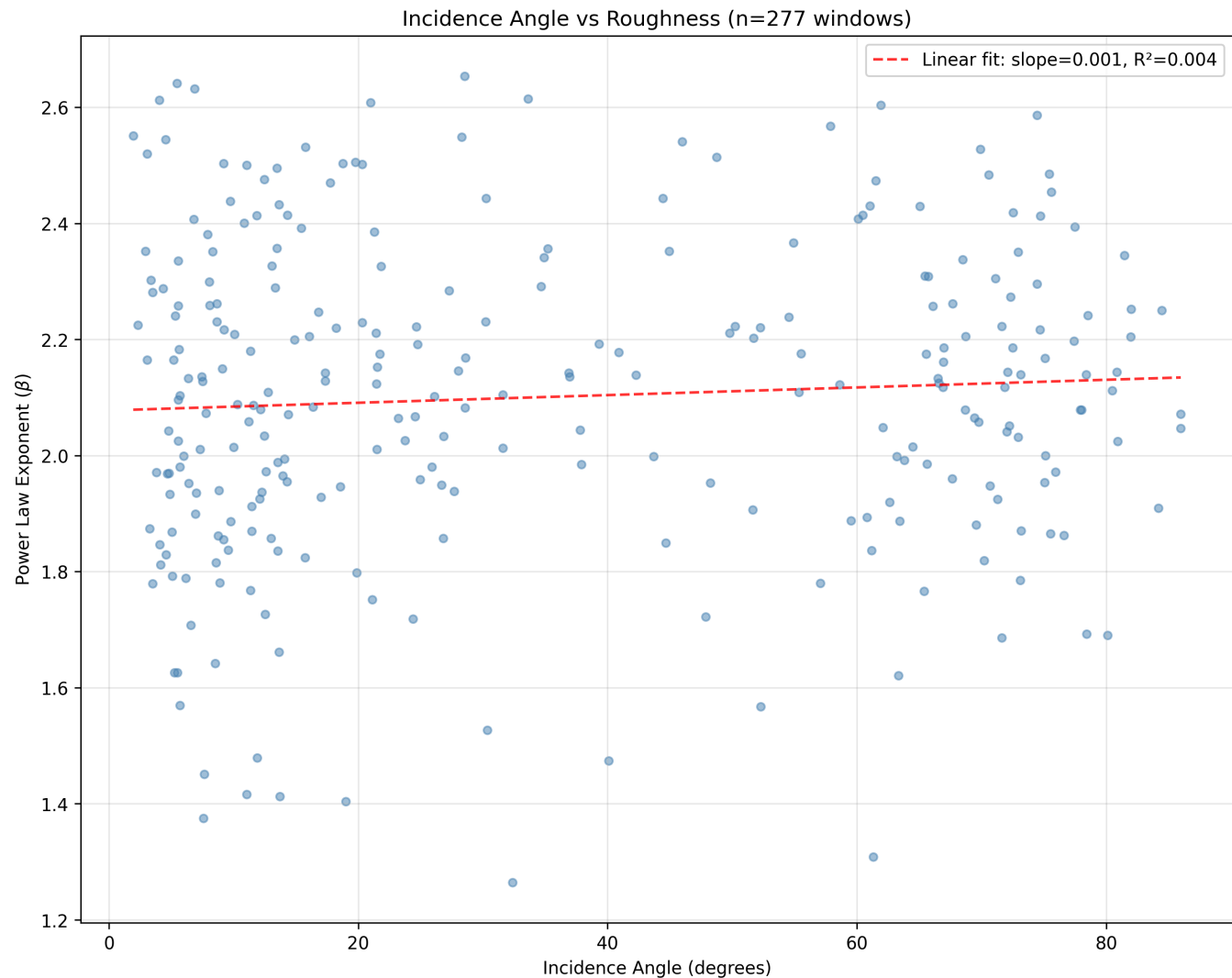
This is below 1.96 (95% confidence interval), which means that the result is not significant.

The $R^2=0.009$ implies that the model explains only 0.9% of variance. **Essentially no directional dependence detected.** This provides support for the **alpine** landscape classification (**isotropic landscape**) and rules out **selective erosion**.

The linear fit at the segment level has **slope=0.0008**, **$R^2=0.006$** , **$p=6.35e-01$**

Since **$p < 0.05$** means significant. At 0.635 this result is non-significant (there's a 64% chance you'd see this slope (or steeper) from pure noise). This means that the slope is essentially zero.

Window-level Roughness FIT



The window level analysis show a near-flat line fit (small slope, very modest R^2). That means that there is no systematic dependence of roughness on flow orientation, this is consistent with a non-streamlined landscape providing support for the **Alpine landscape** classification.

Conclusion:

There is no substantial impact in the anisotropy (isotropy) result from changing the window size to 30km.

75 km

Terminal

```
==== POLARGAP_2015_Fig1_Pensacola_Pole SUMMARY ====
-----
RESULTS SUMMARY (20 trajectories aggregated)
-----
VERTICAL RELIEF (Max-Min):
-> Mean: 1311.3m | Range: [442.2, 2437.3]m
```

```

ABSOLUTE BED ELEVATION:
-> Min: -1874.8m | Max: 1225.2m (across all trajectories)
AVG SEGMENT LENGTH:
-> Mean: 114066.9m | Range: [20970.2, 262973.5]m
POWER LAW EXPONENT (Beta):
-> Mean: 2.2 | Range: [1.3, 2.8]
HURST EXPONENT:
-> Mean: 0.6 | Range: [0.2, 0.9]
MEAN ICE THICKNESS:
-> Mean: 2990.6m | Range: [1801.3, 3821.0]m
.....
FLOW ORIENTATION (All Segments):
-> Perpendicular: 14 segments (36.8%)
-> Parallel: 18 segments (47.4%)
-> Oblique: 6 segments (15.8%)
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
-> Count: 1764
-> Mean: 1228.0m | Range: [54.2, 37289.6]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
-> Count: 4
-> Range: [10777m, 13772m]
.....
=====
Exported 181 window rows and 38 segment rows

```

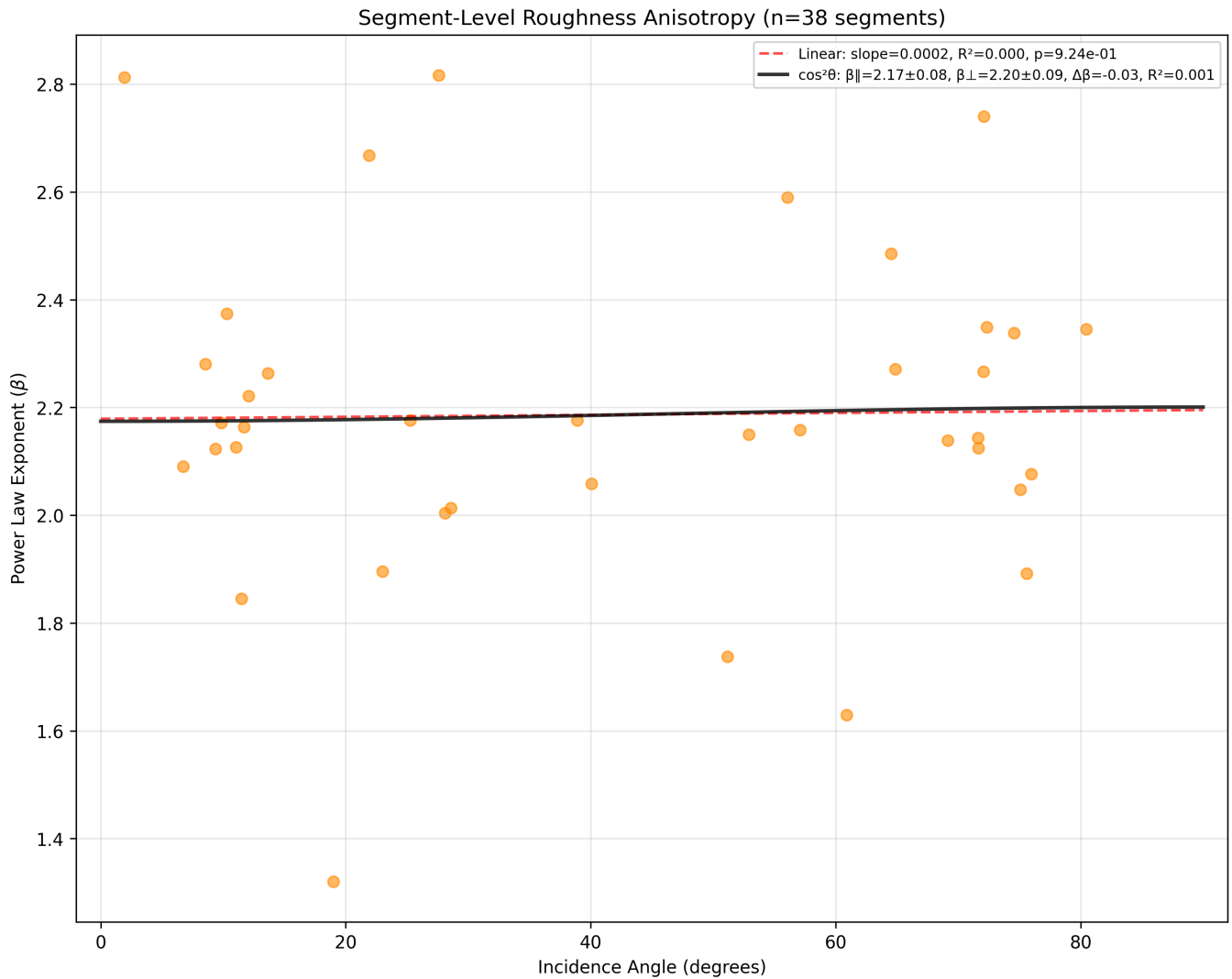
Observations

Mean β value and range, mean vertical relief and range, aggregated height minimum and maximum values are all IDENTICAL to the 50km window condition.

The mean confirmed wavelength of 1228.0 m with the range going up to ~37 km indicates the roughness is dominated by structures at roughly the km-scale.

Anisotropy

Segment-level Roughness FIT



The $\cos^2 \theta$ fit: $\beta_{\parallel} = 2.17 \pm 0.08$, $\beta_{\perp} = 2.20 \pm 0.09$, $\Delta\beta = -0.03$, $R^2=0.001$

The error bars of the parallel and perpendicular beta parameters **overlap** substantially:

2.13 ± 0.07 gives $[2.09, 2.25]$, and 2.20 ± 0.09 gives $[2.11, 2.29]$, this means that parallel and perpendicular roughness are indistinguishable

The anisotropy signal $\Delta\beta = -0.03$

Its significance can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.03}{\sqrt{0.08^2 + 0.09^2}} \approx \frac{0.03}{0.120} \approx 0.25$

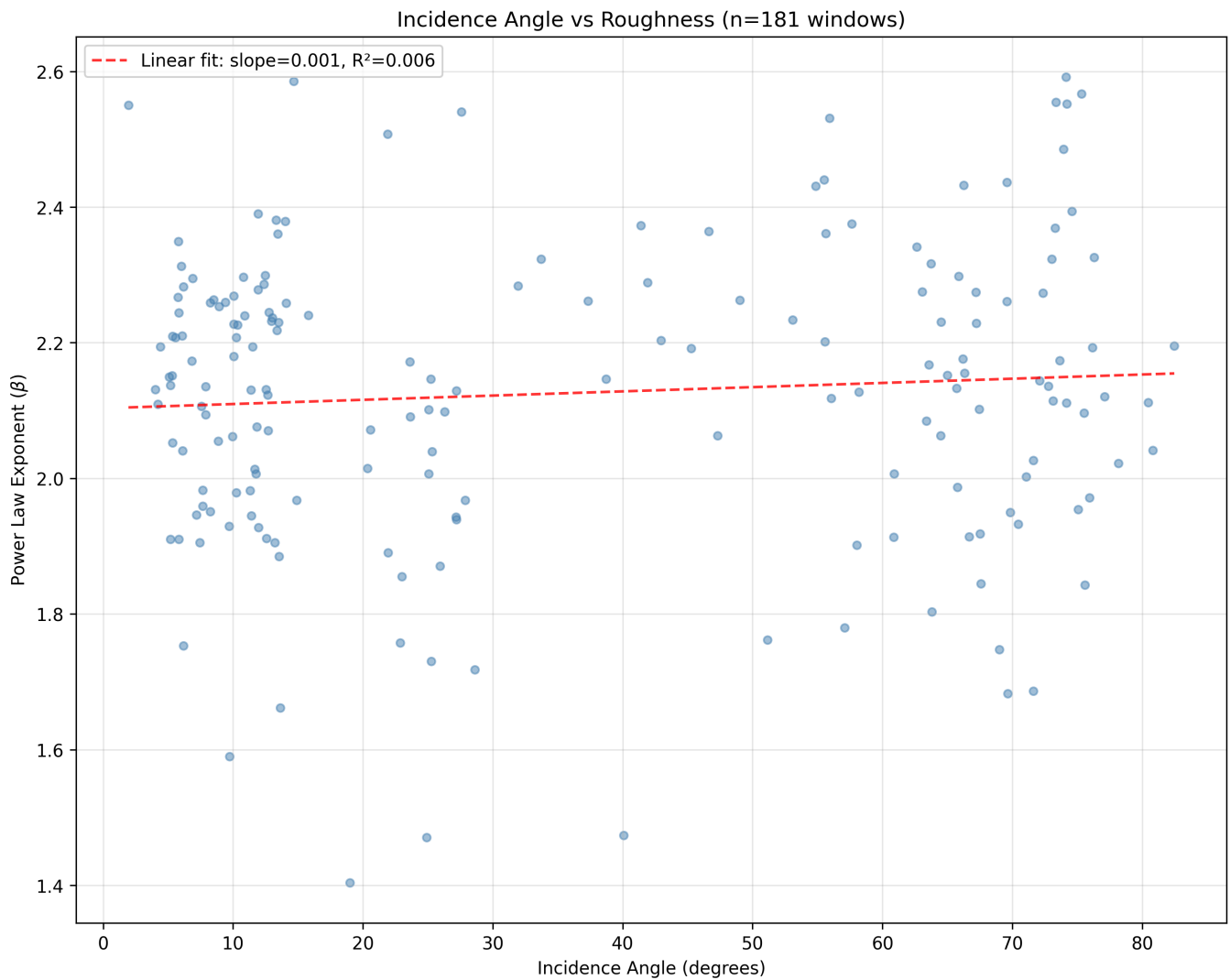
This is below 1.96 (95% confidence interval), which means that the result is not significant.

The $R^2=0.001$ implies that the model explains only 0.1% of variance. **Essentially no directional dependence detected.** This provides support for the **alpine** landscape classification (**isotropic landscape**) and rules out **selective erosion**.

The linear fit at the segment level has **slope=0.0002**, **$R^2=0.000$** , **$p=9.24e-01$**

Since **$p < 0.05$** means significant. At 0.924 ($p > 0.05$) this result is non-significant (there's a 92% chance you'd see this slope (or steeper) from pure noise). This means that the slope is essentially zero.

Window-level Roughness FIT



The window level analysis show a near-flat line fit (small slope, very modest R^2). That means that there is no systematic dependence of roughness on flow orientation, this is consistent with a non-streamlined landscape providing support for the **Alpine landscape** classification.

Conclusion:

There is no substantial impact in the anisotropy (isotropy) result from changing the window size to 75 km.

100 km

Terminal

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=== POLARGAP_2015_Fig1_Pensacola_Pole SUMMARY ===
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-----
RESULTS SUMMARY (20 trajectories aggregated)
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VERTICAL RELIEF (Max-Min):
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```
-> Mean: 1311.3m | Range: [442.2, 2437.3]m
```

```
ABSOLUTE BED ELEVATION:
```

```

-> Min: -1874.8m | Max: 1225.2m (across all trajectories)
AVG SEGMENT LENGTH:
-> Mean: 114066.9m | Range: [20970.2, 262973.5]m
POWER LAW EXPONENT (Beta):
-> Mean: 2.2 | Range: [1.3, 2.8]
HURST EXPONENT:
-> Mean: 0.6 | Range: [0.2, 0.9]
MEAN ICE THICKNESS:
-> Mean: 2990.6m | Range: [1801.3, 3821.0]m
.....
FLOW ORIENTATION (All Segments):
-> Oblique: 6 segments (15.8%)
-> Perpendicular: 14 segments (36.8%)
-> Parallel: 18 segments (47.4%)
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
-> Count: 1908
-> Mean: 1405.3m | Range: [54.2, 57529.3]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
-> Count: 4
-> Range: [10777m, 13772m]
.....
=====
Exported 141 window rows and 38 segment rows

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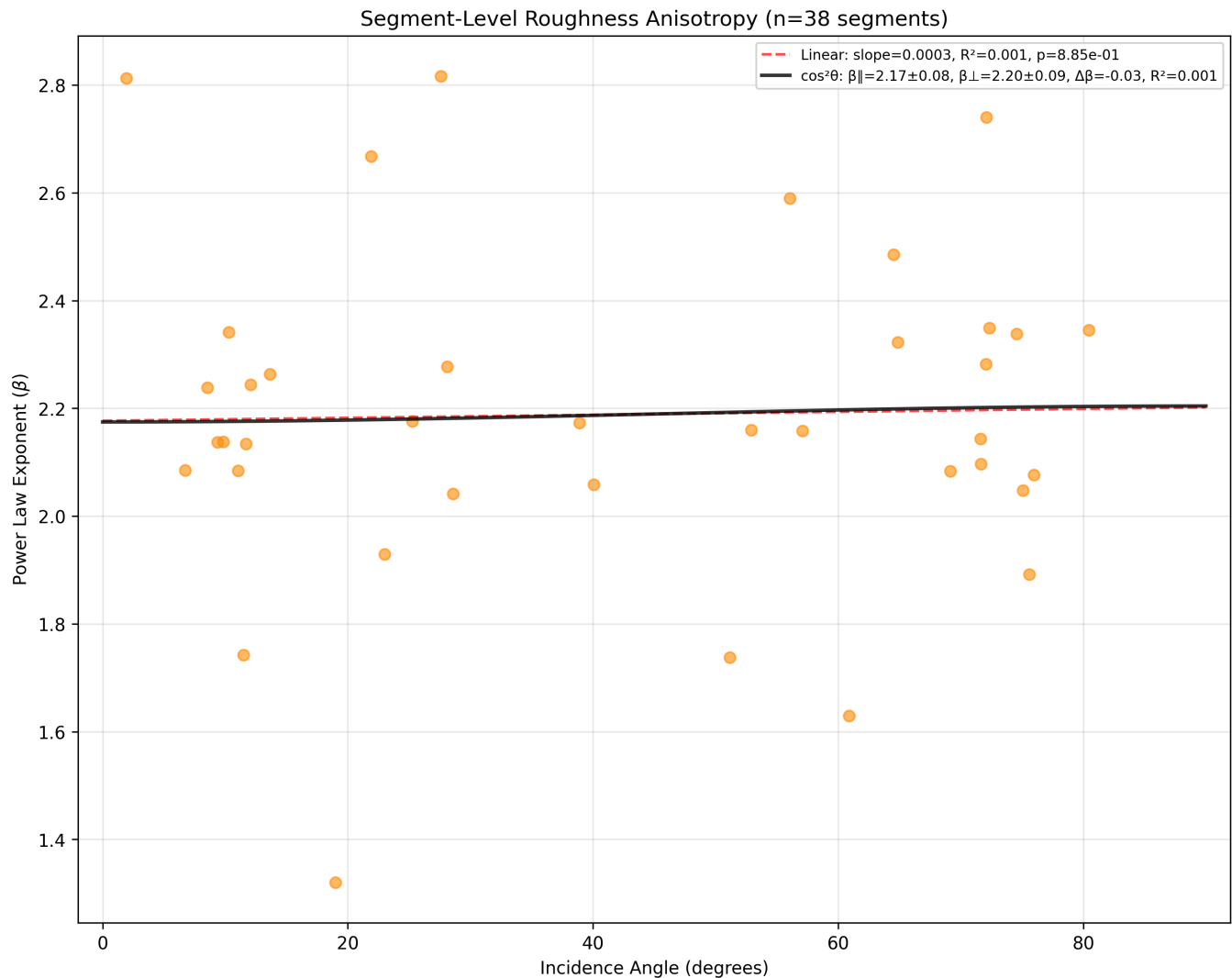
Observations

Mean β value and range, mean vertical relief and range, aggregated height minimum and maximum values are all IDENTICAL to the 50km window condition.

The mean confirmed wavelength of 1405.3 m with the range going up to ~58 km indicates the roughness is dominated by structures at roughly the km-scale.

Anisotropy

Segment-level Roughness FIT



The $\cos^2\theta$ fit: $\beta_{\parallel} = 2.17 \pm 0.08$, $\beta_{\perp} = 2.20 \pm 0.09$, $\Delta\beta = -0.03$, $R^2=0.001$

The error bars of the parallel and perpendicular beta parameters **overlap** substantially:

2.17 ± 0.08 gives [2.09, 2.25], and 2.20 ± 0.09 gives [2.11, 2.29], this means that parallel and perpendicular roughness are indistinguishable

The anisotropy signal $\Delta\beta = -0.03$

Its significance can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.03}{\sqrt{0.08^2 + 0.09^2}} \approx \frac{0.03}{0.120} \approx 0.24$

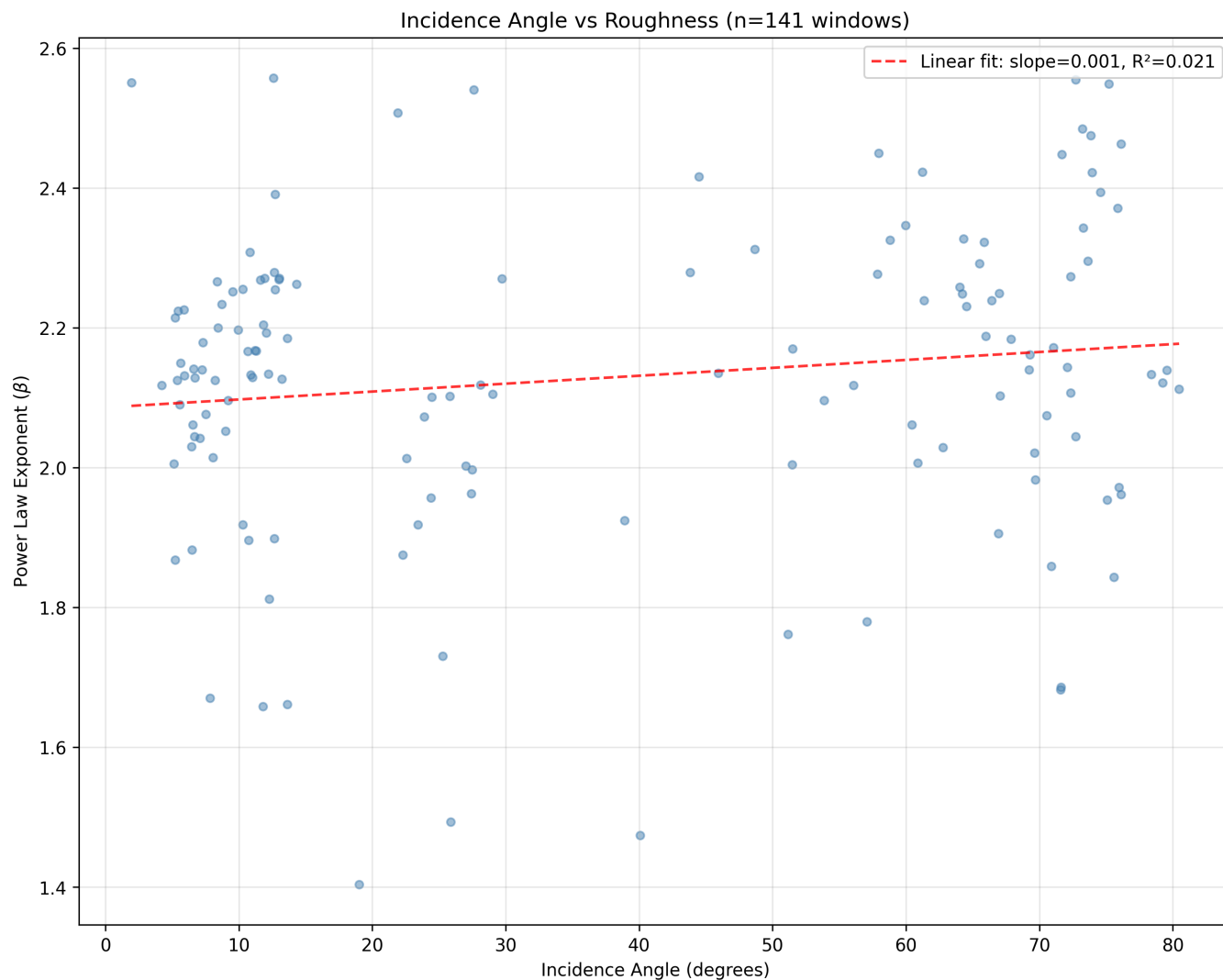
This is below 1.96 (95% confidence interval), which means that the result is not significant.

The $R^2=0.001$ implies that the model explains only 0.1% of variance. **Essentially no directional dependence detected.** This provides support for the **alpine** landscape classification (**isotropic landscape**) and rules out **selective erosion**.

The linear fit at the segment level has **slope=0.0003**, **$R^2=0.001$** , **$p=8.85e-01$**

Since **$p < 0.05$** means significant. At 0.885 this result is non-significant (there's a 89% chance you'd see this slope (or steeper) from pure noise). This means that the slope is essentially zero.

Window-level Roughness FIT



The window level analysis show a near-flat line fit (small slope, very modest R^2). That means that there is no systematic dependence of roughness on flow orientation, this is consistent with a non-streamlined landscape providing support for the **Alpine landscape** classification.

Conclusion:

No substantial difference in results is observed based on window size change to 100 km.